CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

CANDIDATE NAME		CLASS	2T

INDEX NUMBER				

BIOLOGY

9744/04

Paper 4 PRACTICAL

14th AUGUST 2017
2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Index number, name and class on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
TOTAL	

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

1 Investigation into the effect of changing the concentration of an enzyme on enzyme activity.

The biological molecule, U, reacts with water to form aqueous ammonium carbonate. The enzyme urease catalyses this reaction.

Aqueous ammonium carbonate produces ammonium ions. These form an alkaline solution which causes red litmus paper to turn blue. The time taken for red litmus paper to turn blue can be used to monitor the progress of the reaction.

You are required to investigate the effect of enzyme concentration on this reaction.

You are provided with the following:

- 15 cm^3 of 10.0% urease solution, **E**, which is an irritant
- 100 cm^3 of distilled water, **W**
- 25 cm^3 of a solution of the biological molecule, **U**
- Red litmus paper, total length of about 20 cm

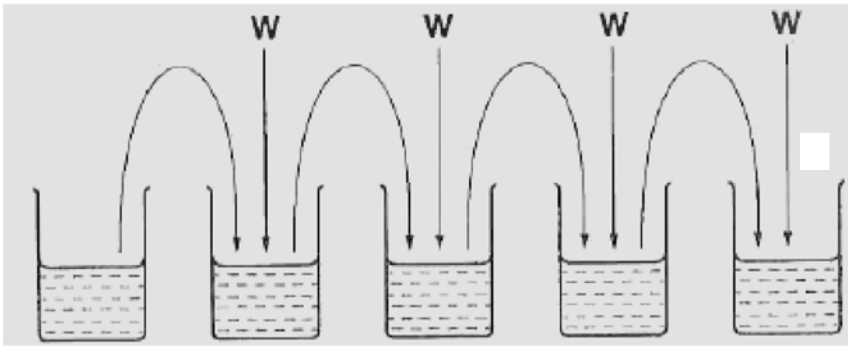
It is recommended that you wear safety goggles / glasses.

1 Carry out a serial solution of the urease solution, **E**, to reduce the concentration of the enzyme by half between each of the four successive dilutions, and set up a control.

Label four small beakers, **D1, D2, D3** and **D4**, for the serial dilutions and label another small beaker, **C**, for the control.

Complete Table 1.1 to show how you will make the different concentrations of urease solution and how you will set up the control, **C**.

Table 1.1



Solution	**E**	**D1**	**D2**	**D3**	**D4**
concentration of urease / %					
volume of urease solution to be diluted / cm^3					
Volume of distilled water, **W** / cm^3					

description of the control, **C:**

....

....

[3]

2 In order to monitor the progress of the reaction, in step **4** red litmus paper will be added to each mixture of enzyme (urease) and substrate, **U**, in a test-tube. To prevent the paper sticking to the wall of the test-tube, you will need to use the glass rod to add it, as follows.

Cut a piece of red litmus paper so that it is a little shorter than the circumference of the glass rod. Moisten the paper and stick it to the end of the glass rod as shown in Fig. 1.1. The glass rod can then be lowered into the mixture of urease enzyme and substrate, **U**. The red litmus paper will slip off into the mixture and the glass rod can be removed.

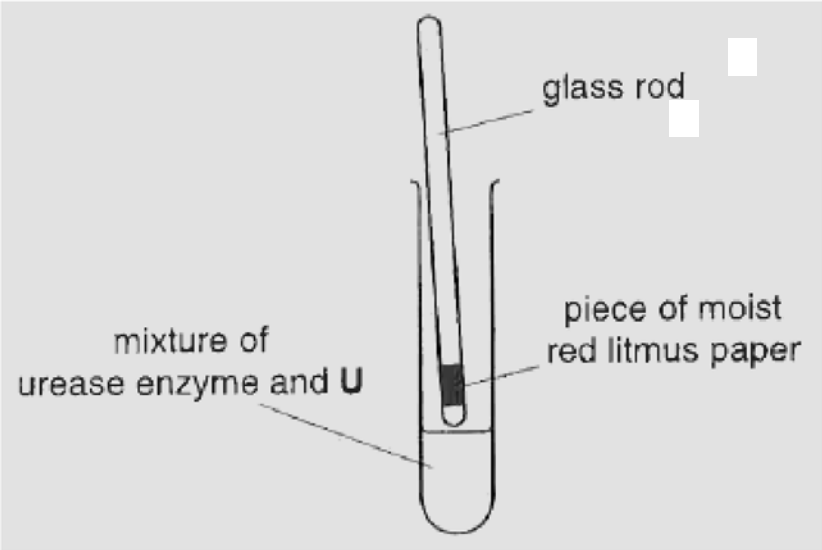


Fig. 1.1

3 Prepare a table in the space on page 4 (step **7**) to record the results of this investigation at various concentrations of urease solutions, including the control.

Proceed as follows:

4 To test the activity of the highest concentration of urease solution, put 2 cm^3 of the substrate, **U**, into a test-tube then add 2 cm^3 of **E** and mix well. The reaction will start as soon as **E** is added.
Immediately, put one piece of red litmus paper into the test0tube as described in step **2** and start timing.

5 Record, in the table that you have prepared on page 4 (step **7**), the time taken for the piece of red litmus paper to turn blue. If the piece of red litmus paper does not turn blue in ten minutes, record 'more than 600'.

6 Record steps **4** and **5** for the other concentrations of urease solution, **D1, D2, D3** and **D4**, and the control, **C**. The red litmus paper used each time should be of the same size.

7 Use the space below to record your results.

[3]

8 Calculate the rate of reaction, using your result for the 10.0% concentration of the urease solution, **E**.

rate of reaction .... [1]

9 Lack of repeats is one limitation of this procedure. Describe one significant source of error in this procedure that also acts as a limitation.

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.... [1]

10 Suggest how you would make **one** improvement to this procedure to reduce the effect of the significant source of error identified in step **9**.

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.... [1]

The effect of pH on the activity of two proteolytic enzymes, **A** and **B**, was compared. The substrate for the enzyme was coloured jelly, which is made of protein.

The apparatus of each pH was set up as shown in Fig. 1.2.

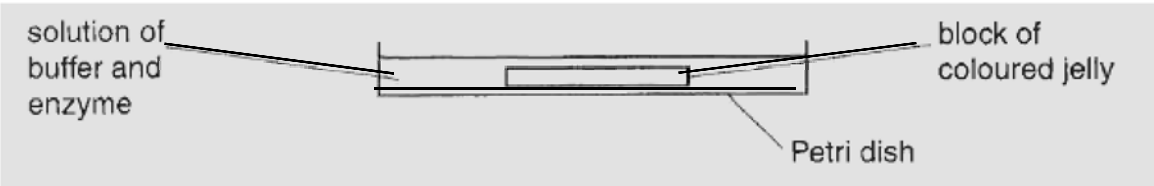

Fig. 1.2

The block of coloured jelly get smaller as it is digested by the enzymes.

11 State two variables which would need to be controlled. Suggest how each variable would be controlled.

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.... [3]

The results of the investigation are shown in Table 1.2.

Table 1.2

pH	area of jelly present after 90 minutes / mm^2	
	enzyme **A**	enzyme **B**
4.0	10	134
6.4	76	124
7.4	128	76
8.0	138	52
9.0	140	6

12 Plot, on the grid opposite, the data shown in Table 1.2. Draw lines of best fit for enzyme **A** and enzyme **B**.
[4]

13 **(a)** Describe the effect of pH on the activity of enzymes **A** and **B**.

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.... [1]

(b) Suggest and explain why changes in pH affect the activity of these two enzymes differently.

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.... [3]

[Total: 20]

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2 Stomata, in the epidermis of leaves, are responsible for the exchange of gases and the release of water vapour. A pair of guard cells controls the opening and closing of each stoma. In the guard cell membrane there is a transport protein. During the opening of the stomata, this protein uses energy from the hydrolysis of ATP to move protons (H^+) out of guard cells. This has two effects.

- Because protons are positively charged, their removal from the guard cells causes the interior of the cells to become negatively charged relative to the exterior. Because of this, some positive ions such as K^+ move into the interior of the cells lowering the water potential inside the cells.
- The pH inside the cell is increased.

You are provided with leaf samples that are soaking in the following solutions.
solution **X**: 0.1 mol dm^{-3} potassium chloride at pH 7.0
solution **Y**: 0.1 mol dm^{-3} sodium chloride at pH 7.0
solution **Z**: 0.1 mol dm^{-3} potassium chloride at pH 4.5

Proceed as follows:

1 Use a pair of forceps to remove the leaf from solution **X** and use scissors to cut out an area up to 1 cm × 1 cm. Transfer this to a slide ensuring **that the lower epidermis is uppermost**. Use a dropping pipette to add a drop or two of solution **X** to the leaf surface. Lower a cover slip over the leaf being careful to exclude any air bubbles.

2 Use the 10X objective of a microscope to locate the stomata. Count the **total** number of stomata that are visible and the total number that are fully **open** in the same field of view. Ignore any that you are doubtful about. Repeat this for another two areas of the leaf. Calculate the mean percentage of open stomata.

3 Repeat steps **1** and **2** for leaves from solutions **Y** and **Z** using clean slides and cover slips on each occasion. You **must** keep slide **Z** in order to answer **(b)** on page **11.**

(a) **(i)** Record your results in an appropriate format in the space provided below.

[4]

(ii) Explain your results.

X ....
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Y ....
[2]

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[2]

Z ....
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[2]

(iii) Guard cells, unlike other cells in the epidermis, have chloroplasts. These chloroplasts have grana but lack the enzymes necessary for the light-independent stage of photosynthesis (Calvin cycle).

With reference to your results, suggest why guard cells have chloroplasts if they do not carry out the light-independent stage of photosynthesis.

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[2]

(b) Make a **high power** drawing of two guard cells and the epidermal cells on either side of each guard cell from the leaf in solution **Z.**

[3]

(c) Fig. 2.1 shows a diagram of a stage micrometer scale that is being used to calibrate an eyepiece graticule.

One division, on either the stage micrometer scale or the eyepiece graticule, is the distance between two adjacent lines.

The length of one division on this stage micrometer is **0.1 mm**.

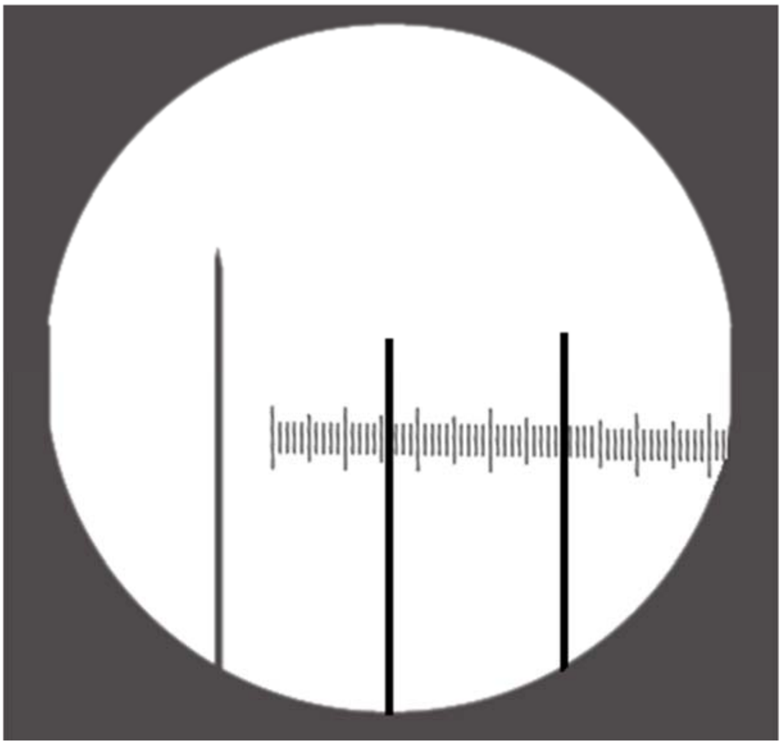

Fig. 2.1

(i) Using this stage micrometer, where one division is 0.1 mm, calculate the actual length of one eyepiece graticule divison, using Fig. 2.1.

Convert your answer to a measurement units most suitable for use in light microscopy. Show the steps and units in your calculation.

[2]

Fig. 2.2 shows a photomicrograph of plant cells some of which have lost water by osmosis.

Fig 2.2

A student, using a prepared slide from which this photomicrograph was taken, measured the total length of the seven chloroplasts, labelled in **cell Z** in Fig 2.1.

Fig. 2.3 shows the view that the student saw when using the eyepiece graticule, calibrated in **c(i)** at the high-power of a microscope.

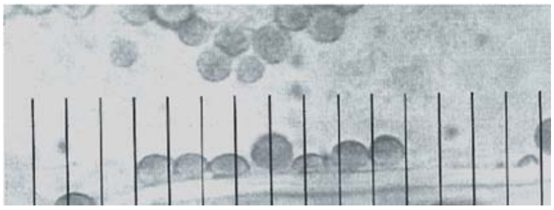

Fig 2.3

(ii) Using this and the information in **c(i),** calculate the actual mean length of one chloroplast as shown in Fig 2.3.

Show the steps and units in your calculation.

actual mean length of one chloroplast .... [2]

(d) Fig. 2.4 and Fig. 2.5 are photomicrographs of the lower surface of the leaf from two different plants, with the same field of view, using the same objective lens.

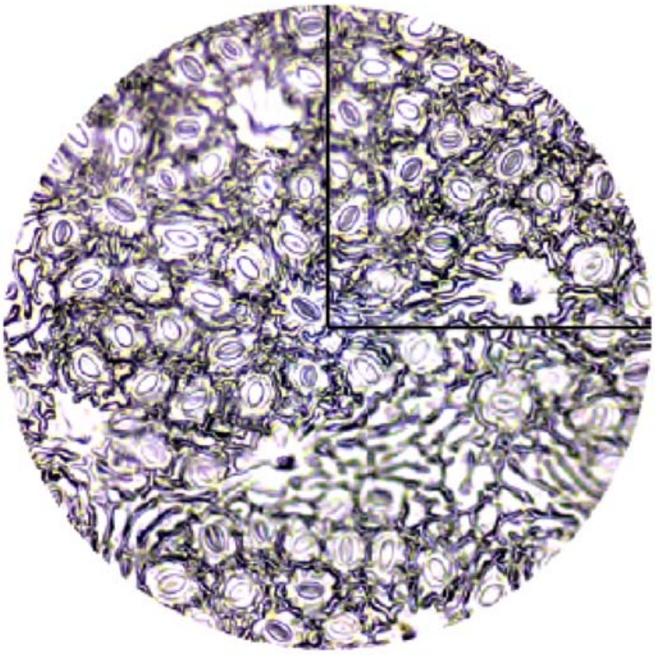

Fig. 2.4

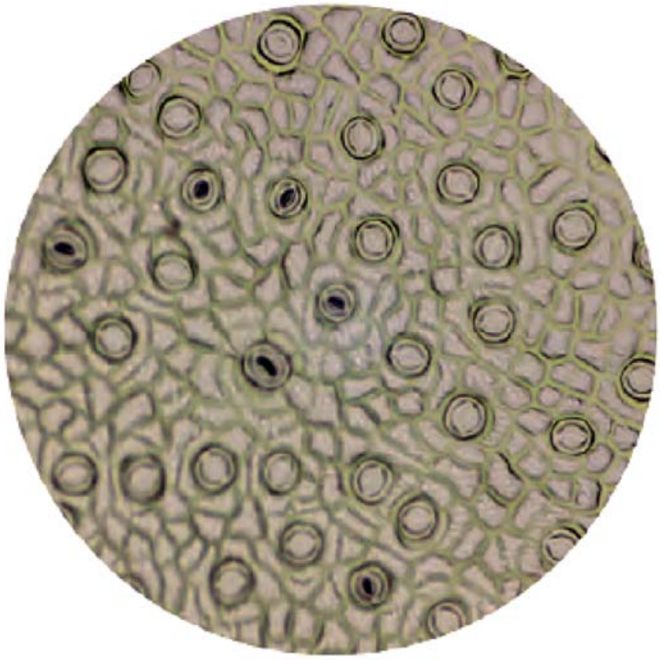

Fig. 2.5

Complete the table below to record 2 observable differences between the surface of each leaf shown in Fig. 2.4 and Fig. 2.5.

Feature	Fig. 2.4	Fig. 2.5

[2]

[Total: 22]

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3 The enzyme urease is a catalyst of the hydrolysis of urea in solution, forming ammonia and carbon dioxide, for example in the breakdown of urea in soils by microorganisms.

You are required to plan an investigation to compare the activity of urease free in solution and urease immobilised in alginate beads.

As the reaction proceeds, the ammonia released dissolves, causing the pH to increase.

You are provided with the following equipment which you may use or not in your plan, as you wish. You may **not** use any additional equipment in your plan.

- an unlimited supply of calcium alginate beads, all of uniform size, prepared with a 50 g dm^{-3} urease solution (you may call this immobilised urease)
- an unlimited volume of 50 g dm^{-3} urease solution (you may call this free urease)
- an unlimited volume of 1.0 mol dm^{-3} urea solution
- an unlimited volume of distilled water
- beakers and flasks of different sizes
- stopwatch
- broad and narrow range of pH papers and liquids with appropriate colour charts, pH probes and meters
- colorimeter and tubes/cuvettes
- thermometer
- thermostatically-controlled water baths
- graduated pipettes and pipette fillers
- filter funnels
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube racks

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- include a clear statement of the hypothesis or prediction
- identify the independent and dependent variables
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- describe the method with details and explanations of the procedures that you would adopt to ensure that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment

[Total: 14]

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END OF PAPER